

Customer Lifetime Value Prediction

CRISP-DM Portfolio Report

Customers

38,753

Selected Model

Linear

Test R^2

0.8478

Business objective

Estimate Customer Lifetime Value using only information available at the first purchase, supporting acquisition-budget allocation, product and channel prioritization, segmentation, and CAC decisions.

Test-set performance

- R^2 0.8478
- RMSE R\$ 507.74
- MAE R\$ 404.03
- 41 transformed features

38,753 customers | Historical LTV represented: R\$ 84,964,816.22

All figures in this report come from the executed Python workflow and the prepared modeling dataset.

Model Evaluation

Five-fold cross-validation on the training set and held-out test performance

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean R ²
Dummy	-0.0001	-0.0026	-0.0011	-0.0000	-0.0004	-0.0009
Linear	0.8489	0.8440	0.8533	0.8486	0.8572	0.8504
Poly (d=2)	0.8434	0.8399	0.8501	0.8455	0.8533	0.8464
RF	0.8210	0.8152	0.8313	0.8254	0.8339	0.8254

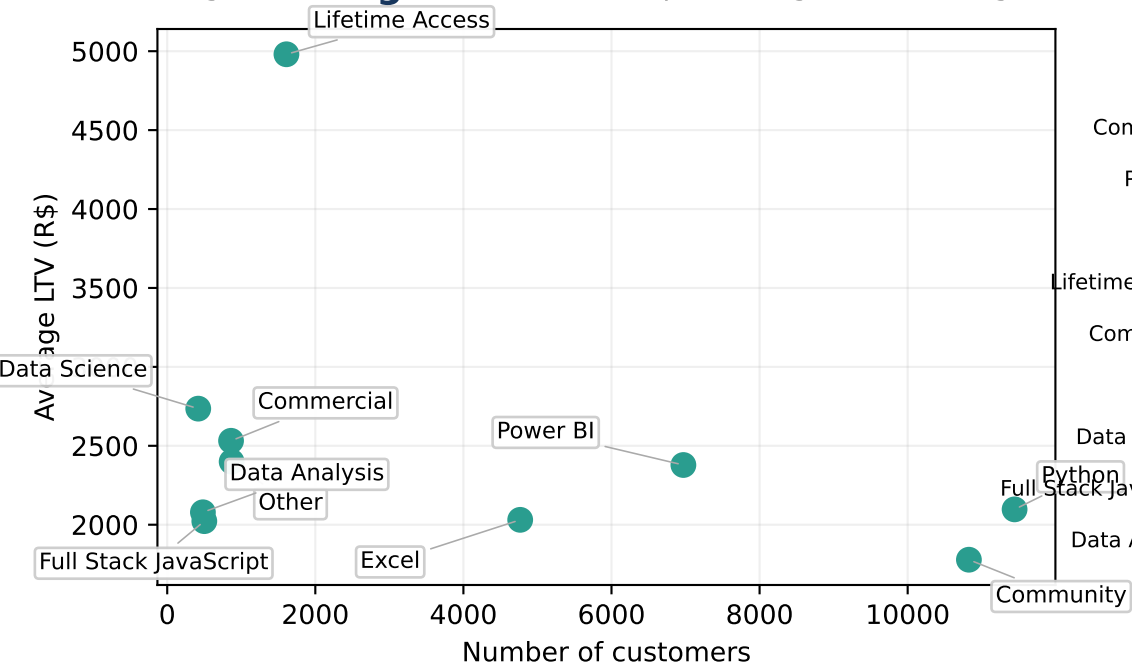
Model	Test R ²	RMSE	MAE
Dummy	-0.0001	R\$ 1,301.31	R\$ 1,024.58
Linear	0.8478	R\$ 507.74	R\$ 404.03
Poly (d=2)	0.8453	R\$ 511.88	R\$ 407.41
RF	0.8229	R\$ 547.69	R\$ 435.37

Linear Regression was selected: strongest validation performance with high interpretability.

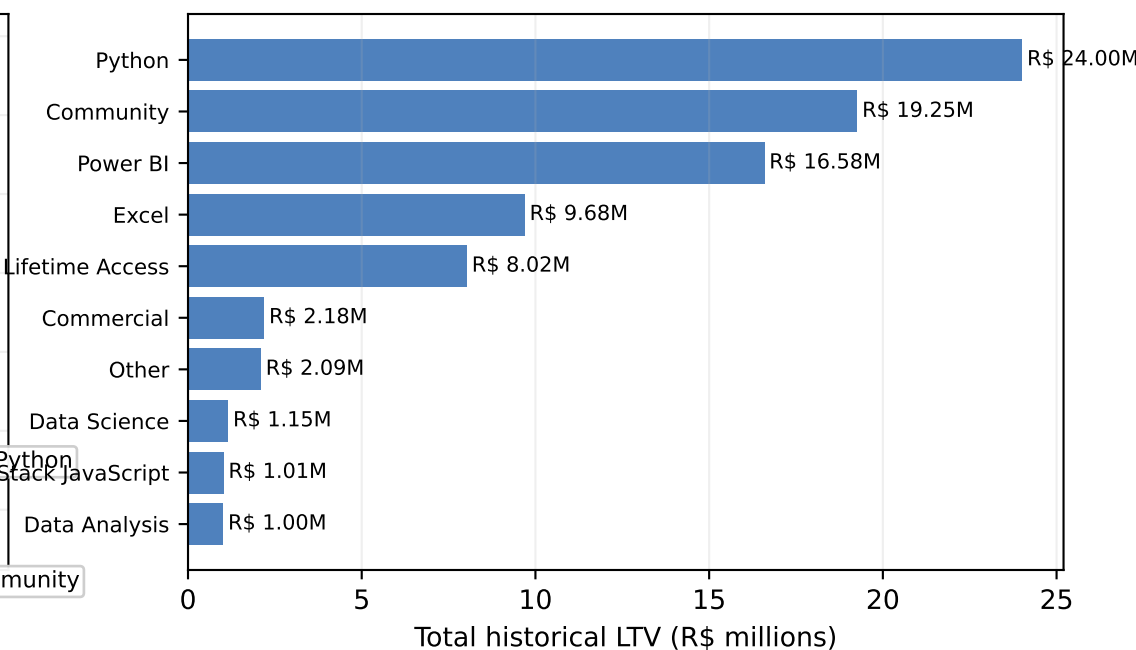
Entry Product Analysis

Average customer LTV must be high enough to support segment scale

Average LTV vs Customer Volume



Total Historical LTV

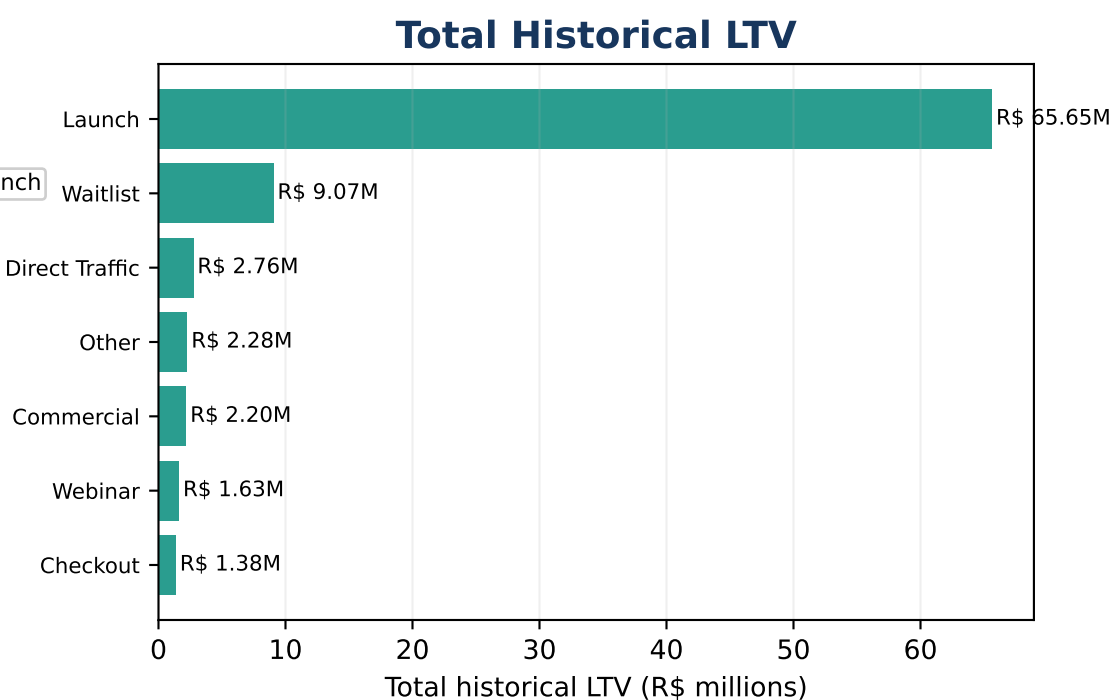
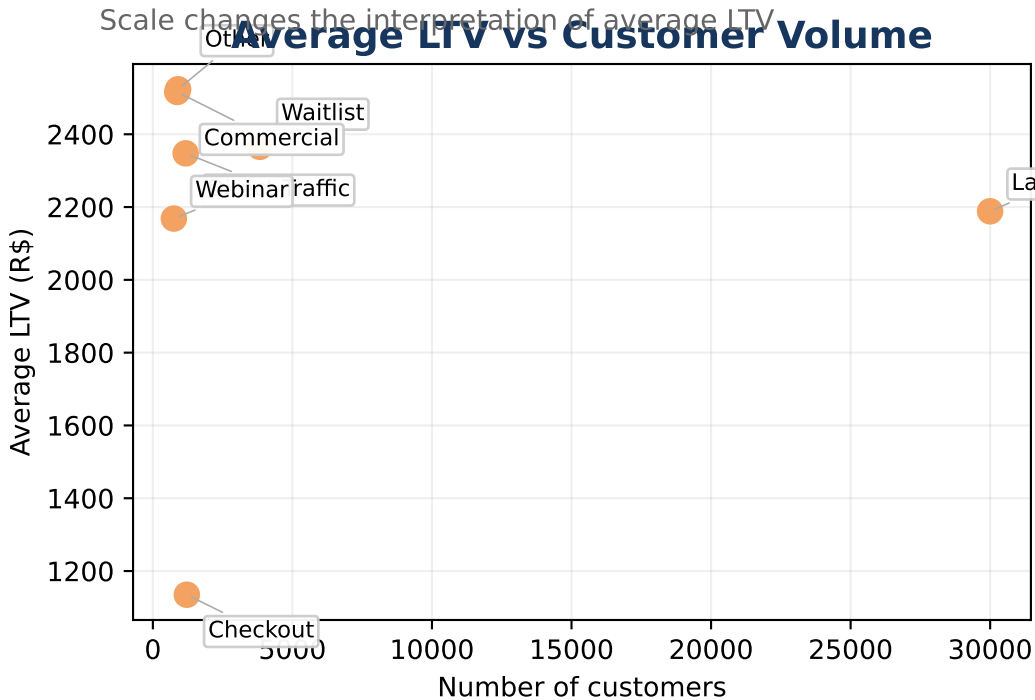


Entry Product	Customers	Share	Average LTV	Total LTV	LTV Share
Python	11,444	29.53%	R\$ 2,097.22	R\$ 24,000,592.46	28.25%
Community	10,827	27.94%	R\$ 1,778.27	R\$ 19,253,351.38	22.66%
Power BI	6,971	17.99%	R\$ 2,378.33	R\$ 16,579,320.09	19.51%
Excel	4,767	12.30%	R\$ 2,030.97	R\$ 9,681,629.90	11.39%
Lifetime Access	1,610	4.15%	R\$ 4,980.48	R\$ 8,018,576.27	9.44%
Commercial	862	2.22%	R\$ 2,531.73	R\$ 2,182,350.29	2.57%
Other	870	2.24%	R\$ 2,399.47	R\$ 2,087,536.22	2.46%
Data Science	419	1.08%	R\$ 2,735.36	R\$ 1,146,116.39	1.35%
Full Stack JavaScript	501	1.29%	R\$ 2,022.65	R\$ 1,013,349.42	1.19%
Data Analysis	482	1.24%	R\$ 2,078.83	R\$ 1,001,993.80	1.18%

Key insight: Lifetime Access leads in LTV per customer; Python leads in total historical LTV because of scale.

Sales Channel Analysis

Scale changes the interpretation of average LTV

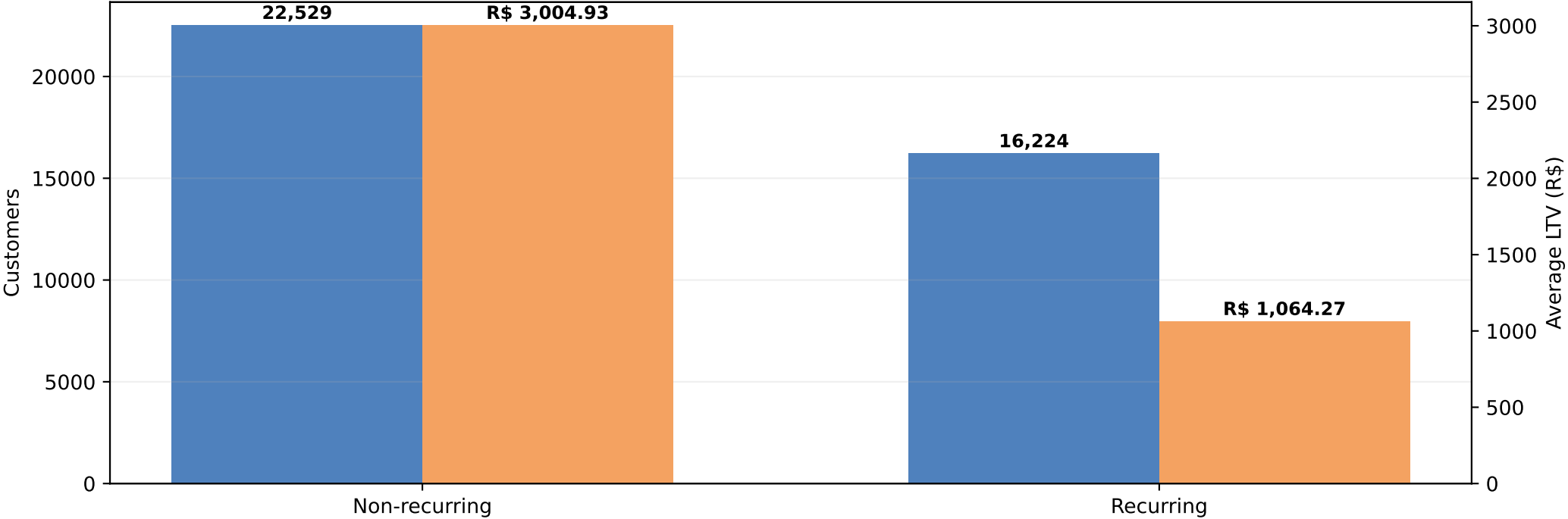


Sales Channel	Customers	Share	Average LTV	Total LTV	LTV Share
Launch	29,998	77.41%	R\$ 2,188.38	R\$ 65,647,040.96	77.26%
Waitlist	3,833	9.89%	R\$ 2,365.55	R\$ 9,067,165.71	10.67%
Direct Traffic	1,176	3.03%	R\$ 2,347.56	R\$ 2,760,735.62	3.25%
Other	904	2.33%	R\$ 2,523.59	R\$ 2,281,326.77	2.69%
Commercial	873	2.25%	R\$ 2,516.30	R\$ 2,196,726.84	2.59%
Webinar	752	1.94%	R\$ 2,168.37	R\$ 1,630,617.80	1.92%
Checkout	1,217	3.14%	R\$ 1,134.92	R\$ 1,381,202.52	1.63%

Key insight: Launch dominates total value because it represents more than three quarters of customers.

Recurring vs Non-Recurring First Purchase

Historical association, not a causal conclusion



Group	Customers	Average LTV	Total LTV	Avg. First Purchase
Non-recurring	22,529	R\$ 3,004.93	R\$ 67,698,074.34	R\$ 1,153.99
Recurring	16,224	R\$ 1,064.27	R\$ 17,266,741.88	R\$ 108.20

Non-recurring customers have much higher observed LTV and first-purchase value. Observation window and LTV definition matter.

Final Business Takeaways

How the model can support acquisition decisions

1

Linear Regression is the selected model

Test R^2 0.8478 | RMSE R507.74|MAER 404.03

2

LTV per customer and segment volume tell different stories

Lifetime Access leads per customer; Python leads in total value.

3

Launch is the economically dominant sales channel

29,998 customers and R\$ 65.65M in historical LTV.

4

Predicted LTV is decision support, not a guaranteed outcome

Combine predicted LTV with CAC, contribution margin, payback, risk and uncertainty.

5

Production improvement: temporal validation

The current project uses a random 80/20 split plus 5-fold CV; temporal validation is the next step.

Deployment artifacts: explained Excel simulator + operational batch scoring workbook.